

ChainLock

Hash-chain lattice memory for aziel-runtime

CL-WP-0.4 9 September 2026 Author: Aziel Eliab / GodLock.AZ Status: local specification. Not a public GitHub release. Not a fleet-completeness claim. Runtime: `chainlock.py` · Schema: `chainlock_schema.json` · Namespace: `chainlock@aziel`

0. Sentence

An AI on this runtime forgets at the session boundary. The lattice does not. ChainLock is the walkable memory between those two facts.

1. Claim

A local set of append-only chains, cross-linked as a lattice and optionally signed with an SSH key, is enough for an MCP-facing model to reconstruct short-term and long-term recollection without an embedding store and without rewriting history.

GodLock already hash-chains a public node. TemporalLock already timestamps receipts. FragGate already appends an act ledger. AZL-WP-1.1 already set UI = MCP. None of those give a reader a depth knob. ChainLock does.

This paper is the doctrine. The spec is `ChainLock_v0_spec.md`. The kit is the runtime. Recognition is not a third kit.

2. What it is not

- Not a public ledger. GodLock remains the public mesh stress-test node at `godlock.uk`.
- Not a substitute for TemporalLock receipts or EmbryoLock vaults. Those stay the evidence and sealed-file tools. ChainLock indexes and walks them.
- Not an embedding store. Match is deterministic: tags, subject, chain, time, hash, fact text.
- Not an untraceable path. SSH stamps identify a local key. Hashes identify bytes.
- Not FoldLock, SweepGate, or AZPIPE. Those are the wire. This is the memory.
- Not Unowned Lattice. UL papers stay UL.
- Not a court filing and not a prophetic document.

3. Units

Stamp. One JSON object. Canonical bytes are UTF-8 JSON with sorted keys and separators (`'`, `,`, `:`). `stamp_sha256` is SHA-256 of that encoding with the hash field omitted. `prev` is the previous stamp's hash on the same chain, or GENESIS for sequence zero.

Card. The wire unit. Slim keys: `id`, `c`, `k`, `t`, `h`, `fh`, `s`, `f`, `g`, `r?`. Extra `facts[]` only on capsules. Subject ≤ 80 characters. Fact ≤ 160 . Cap 4096 bytes. Hash-only cards are non-compliant. Recollection is the fact text. Grounding is `h` (stamp bytes) plus `fh` (SHA-256 of the clipped fact).

Chain. A single hash list stored as jsonl. Default roster: `genesis`, `identity`, `ssh`, `session`, `acts`, `evidence`, `recall`, `mesh`, `library`, `learn`.

Lattice. Typed links between stamps (cites, window-start, born-from, from-interact, quarantine). Links do not mutate the target.

SSH stamp. Detached ssh-keygen -Y sign over the canonical stamp bytes in namespace chainlock@aziel. Fingerprint lives on the identity chain. Unsigned stamps are chain-valid and not identity-grade.

Horizon. stm is the hot window. ltm is retained. both is counted in both. Compaction writes a recall capsule. Capsules hash the ordered member hashes and store the member facts. Source stamps stay.

Depth. Six classes. 0 tip. 1 STM. 2 capsules. 3 expand sources. 4 one lattice hop. 5 genesis or budget. auto starts at STM and deepens until a query hits or the budget dies.

4. Recollection

Short-term: last 16–32 session and acts stamps, plus the live tips.

Long-term: recall capsules first. Expand a capsule only when the question needs a source stamp. Follow one lattice hop only when depth ≥ 4 .

This is the same progressive-disclosure rule the skill loader already uses. Metadata, then body, then the bound file. ChainLock applies it to time.

Every returned fact carries id + h + fh. No claim without a stamp. FragGate door chainlock__door returns a grounded FG-CL envelope or refuse=no-stamp. Do not invent a fact to fill the door.

5. Library tether and adapt

Origin: <https://www.azielcorpuslibrary.net>.

The library chain holds catalog cards only — AZDOC id, title, content_sha256, document chain_tip, receipt URL. PDF bodies stay on the site. vault/library.jsonl is the offline shelf. If the Worker dies or a record is deleted from the public catalog, recall still has the card.

tether binds the current SSH identity to the library window hash. One stamp covers the set. New syncs require a new tether. Do not invent AZDOC ids or file hashes. Sync from /v1/search. Quarantined rows stay off the shelf.

Every operator action or choice gets an interact stamp on session. choose / refuse / -- learn also write learn. Adaptive recall is that chain plus its capsules. This is not a weight file and not a fine-tune.

6. MCP

The operator surface is tools, not a page.

```
chainlock__status chainlock__append chainlock__walk chainlock__recall chainlock__verify
chainlock__compact chainlock__door chainlock__mesh_cite chainlock__library_sync
chainlock__tether chainlock__interact chainlock__seal chainlock__pipe
```

Local process. No public panel. No IP allow/block UI. No Node Gate.

7. LOCKSET

seal writes three things that must agree:

1. Every live chain tip (id, h, fh). 2. A TemporalLock receipt: neutral timestamp, no narrative, no authority. 3. A GodLock cite: <https://godlock.uk>, pull=false, isolate=true.

verify is fail-closed. One broken prev, one receipt hash miss, one missing GodLock cite, or one tip drift after seal returns `lattice:false`. Append, interact, or library-sync after seal drifts tips. Reseal.

This runtime cites GodLock. It does not write the public ledger. The operator posts the lockset SHA-256.

8. Failure and poison

Broken prev or mismatched `stamp_sha256` isolates that chain. Do not rewrite. Append a verify stamp recording the break hash.

Failed SSH verify flags `SSH_FAIL`. The stamp stays.

Poison or unexpected remote citation records on mesh as `rel=quarantine`. SweepGate airlocks the inbound envelope before entry. Remote body does not enter STM.

Phoenix / ARK re-seal of a bound file does not edit the old stamp. Append a new pointer.

Court-sealed is a plane tag. It does not encrypt. Use ARK / EmbryoLock for ciphertext. Do not put legal name, home, passwords, or privileged plaintext in a stamp body.

9. Files

```
vault/
  chains/<name>.jsonl
  cards.jsonl
  library.jsonl
  receipts/LOCKSET.json
  receipts/TL-<prefix>.json
  keys/allowed_signers
  sigs/<stamp-id>.sig
  index.json
```

`index.json` and `cards.jsonl` are derived. Source of truth is the jsonl chains. Private SSH keys do not live on the artifacts FUSE mount.

10. Cap

This does not make the model omniscient. It makes forgetting an explicit depth choice instead of an accident.

Compat reads CL-0.1 through CL-0.4. New stamps are CL-0.4.

Companion papers: AP-WP-0.2 (wire), SG-WP-0.1 (airlock), LS-WP-0.1 (fail-closed seal).